An Intensity-Independent Stereo Registration Method of Push-Broom Hyperspectral Scanner and LiDAR on UAV Platforms

Yanfeng Gu, *Senior Member, IEEE*, Chen Wang, *Graduate Student Member, IEEE*, and Xian Li, *Member, IEEE*

***Abstract*— Unmanned aerial vehicles (UAVs) equipped with hyperspectral scanners and light detection and ranging (LiDAR) can flexibly acquire rich spectral and geometric information about the observation scene. To combine the complementary advantages of multisource data, the stereo registration of hyperspectral images and LiDAR data has become one of the hot topics in the remote sensing community. However, existing research works are more focused on exploiting intensity information from multisource data, which is applicable to data acquired on manned vehicle platforms or satellite platforms. For UAV platforms with poor stability and limited load, the low signal-to-noise ratio of LiDAR data and the complex distortion of push-broom images bring great challenges to stereo registration. Under this circumstance, an intensity-independent stereo registration method is proposed in this article, which is based on the physical model of the integrated system and the sensor detection principles. Specifically, the proposed method utilizes the position and orientation system (POS) to reduce the impact of UAV platform motion on hyperspectral imaging, and projection errors are eliminated by the ray tracing model with aid of LiDAR data. Finally, a virtual ray decomposition model based on geometric features is constructed to realize the stereo registration of hyperspectral images and LiDAR data. Compared with an advanced solution and professional processing software, the proposed method has shown better registration performance on two data of different scenarios.**

***Index Terms*— Light detection and ranging (LiDAR), push-broom hyperspectral scanner, stereo registration, unmanned aerial vehicle (UAV).**

I. Introduction

UNMANNED aerial vehicle (UAV) mapping systems are becoming more and more popular in a variety of applications [1]. These systems can be equipped with a variety of sensors such as optical cameras, hyperspectral scanners, and light detection and ranging (LiDAR) sensors to meet the requirements of different industry fields. As a representative of spectral imaging instruments, hyperspectral scanners obtain hyperspectral images with rich spectral and spatial information, which are widely used in classification tasks [2], [3], [4], [5], change detection [6], precision agriculture [7], [8], [9], and so on. In the process of hyperspectral remote sensing imaging detection, spatial information degradation from 3-D to 2-D of ground objects always occurs, resulting in the loss of elevation information. The development of sensor and UAV technologies makes it possible to integrate multiple sensors on the same UAV platform, which can synchronously obtain multisource remote sensing data. By integrating LiDAR and position and orientation system (POS), a 3-D LiDAR point cloud can be obtained quickly and efficiently. However, hyperspectral images and LiDAR data are heterogeneous; hyperspectral images store their spectral information in matrix format, while LiDAR point clouds store their spatial information in the form of discrete 3-D points. These differences in modalities and resolution make them have nonuniform spatial information. The misalignment of the data seriously affects the accuracy of subsequent applications. Data registration of the two data is mandatory in order to combine the advantages of their complementarities.

Data registration refers to the process of geometric or spatial alignment of two sets of data in the same observation scene [10]. The difference in detection mechanisms makes the registration of hyperspectral images and LiDAR data very challenging. To solve this problem, many research efforts have been conducted, which can be divided into the following two research directions: one is based on image registration and another is based on system calibration.

The methods based on image registration employ LiDAR data to generate intensity maps, thereby avoiding the difference in data dimensions between LiDAR data and hyperspectral images. Registration of remotely sensed images is currently conducted with feature-based methods and intensity-based methods [11]. The feature-based registration methods rely on the common features in the template image and the reference image to solve the transformation parameters. In the research of using point features for LiDAR data and optical image registration, some classic feature extraction algorithms, such as the scale-invariant feature transform (SIFT) [12] and the speeded-up robust features (SURFs) [13], have been used for registration of aerial images and LiDAR images. For hyperspectral images, some researchers exploit feature extraction results from SIFT and SURF to achieve registration of hyperspectral

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The authors are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: xianli@hit.edu.cn).

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images and LiDAR data [14]. The difficulty of these methods is to extract and match common features from multisource data. In contrast, intensity-based methods mainly adopt similarity measures to estimate the degree of correlation between images. Typical similarity measures include normalized cross correlation (NCC) [15], [16], mutual information (MI) [17], [18], and normalized gradient fields (NGFs) [19]. Since the intensity information of the data obtained by different types of sensors is significantly different, which will cause errors in the similarity measurement results, in addition, the methods based on image registration are more suitable for simple image transformations, such as translation, rotation, and radial transformation [20]. For complex nonlinear transformations, a large number of matching points are needed to solve the transformation parameters in these methods, which are difficult to achieve the desired effects in the registration of push-broom images.

The methods based on system calibration aim to align the data obtained by different sensors in the same geographic coordinate system [21]. The positioning of the data obtained by LiDAR and hyperspectral scanner is determined by interior orientation parameters (IOPs) and exterior orientation parameters (EOPs). IOPs of hyperspectral scanners include principal distance, principal point coordinates, and lens distortions. EOPs refer to the position and posture of the sensors at the imaging time. IOPs are usually stable and accurate, which are rigorously calibrated by the manufacturer and provided to the users. EOPs can be calculated indirectly from ground control points (GCPs) or directly measured using POS [22], [23]. To reduce labor costs, more researchers have been dedicated to using POS for the geometric correction of push-broom scanners. Some studies establish physical and sensor models to obtain the mapping relationship between scenes and object coordinates [24]. Suomalainen et al. [25] developed a hyperspectral scanner UAV system with an optical frame camera to better utilize the perspective relationship and improve the quality of geometric correction. These methods require rigorous integration of components on the UAV platform to obtain accurate position and attitude information. However, the EOPs obtained by POS need to be calibrated to eliminate the influence of the boresight angles [26].

Extensive research has been conducted to solve the calibration problem on different acquisition platforms. Object-based calibration is a hot research topic in robotics applications. Some scholars take the checkerboard as the target and optimize the parameters by minimizing the distance between the LiDAR points and the image plane [27], [28], [29]. These methods are generally applicable to indoor fixed platforms. In this case, calibration parameters are considered to be fixed values. For airborne platforms and UAV platforms, most works are based on data acquired in flight. The calibration methods of frame imaging cameras and LiDARs are often based on the overlap between images for beam adjustment or 3-D reconstruction. The parameters are calibrated by minimizing the distance between the photogrammetry and LiDAR point clouds [30]. For push-broom scanners, Muller et al. [31] introduced a method for boresight angles' calibration using the coordinates of GCP and projection points. Lenz et al. [32] utilized SURF to detect tie points in overlapping scenes and obtain calibration results by minimizing the projected coordinates of corresponding tie points.

In addition, the above methods are all based on the digital elevation models (DEMs) with 25-m horizontal resolution, which are unsuitable for UAV remote sensing with high spatial resolution. Brell et al. [33] proposed a parametric method for the geometric coalignment of Airborne hyperspectral and LiDAR data. In that approach, the calibration process employs SIFT and random sample consensus (RANSAC) [34], [35] to extract and match feature points and uses the intensity information of the same band of LiDAR and hyperspectral data to construct a cost function to optimize the IOPs and EOPs.

In this article, we propose a novel stereo registration method for hyperspectral images and LiDAR data collected by the UAV platform, which can eliminate the geometric distortion of the hyperspectral images and align these two heterogeneous data. The proposed method is based on the collinearity equations to directly georeference the hyperspectral images assisted by GNSS/INS and LiDAR data. Through the analysis of registration errors between the hyperspectral images and LiDAR data, this method consists of the following steps. First, the POS and LiDAR data are preprocessed to improve the position accuracy of the generated digital surface model (DSM) for geometric reference. Second, an orthorectification module based on ray tracing is set up to reduce the projection error of hyperspectral images. Finally, we present a virtual ray decomposition model to eliminate the effect of boresight deviation on data stereo registration. The flowchart of the proposed method is shown in Fig. 1.

The main contributions of this work are given as follows.

1) In the proposed method, the LiDAR data are used to assist in the strict geometric correction of the hyperspectral image in the stereo registration process for the projection error in the imaging process. This registration model based on the physical model of the multisensor integrated system has good generalization and high registration precision, which is suitable for any scene without being affected by the topographic relief.
2) The proposed method does not rely on LiDAR or hyperspectral intensity information and GCPs. Only a small number of geometric features from LiDAR and hyperspectral data are used in the registration process to yield better reliability.
3) The registration process has no loop optimization and does not need to consider convergence problems. Rough registration is also carried out between the strips of hyperspectral images while realizing the registration of push-broom hyperspectral image and LiDAR.

The rest of this article is organized as follows. Section II describes the characteristics of the multisensor integrated system on the UAV platform. Section III introduces the proposed stereo registration method in detail. The experimental results and analysis are shown in Section IV. Finally, the conclusion is drawn in Section V.

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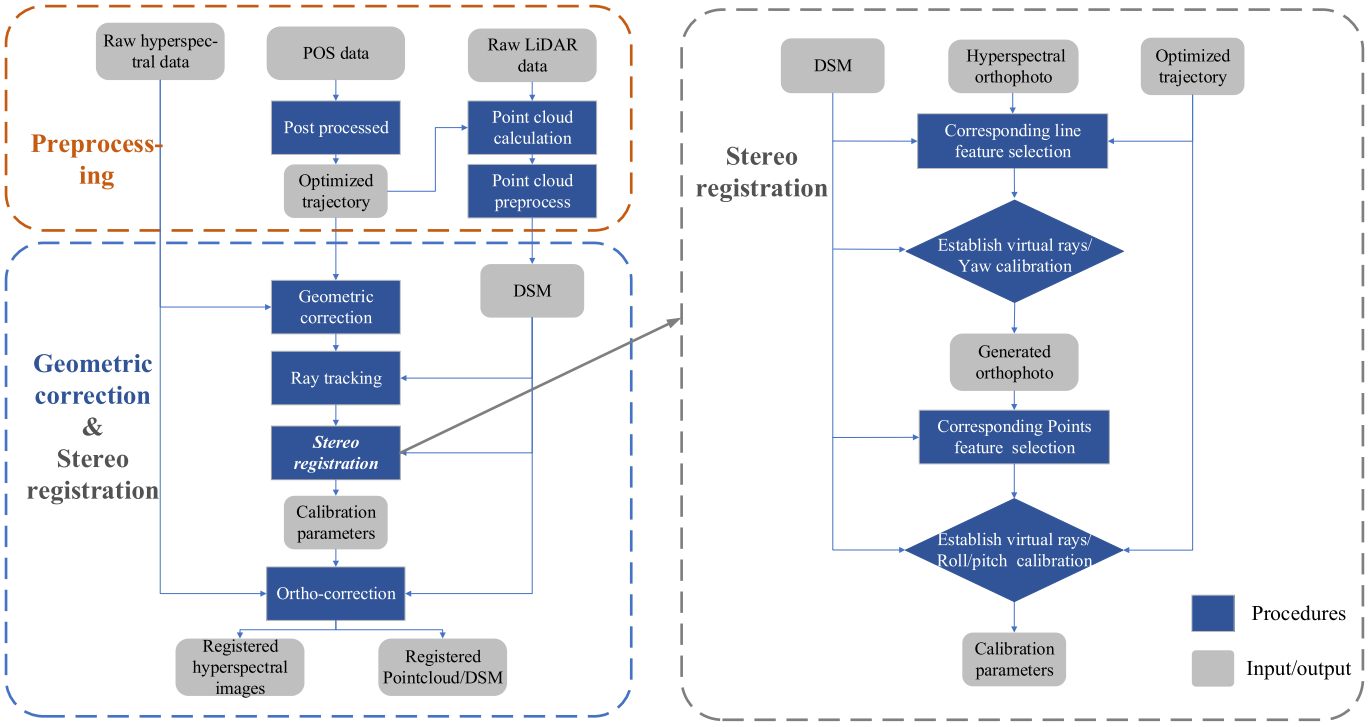


Fig. 1. Flowchart of the proposed method. (Left) Entire processing flow, including data preprocessing, orthorectification, and stereo registration. (Right) Specific steps of stereo registration.

II. Integrated Hyperspectral Scanner and LiDAR System

Hyperspectral scanners and LiDARs are widely used in remote sensing detection. They provide multidimensional surveying and mapping information. With the continuous development of UAV and sensor technologies, hyperspectral scanners and LiDARs can be integrated on the same UAV platform to simultaneously obtain multisource data about the same observation scene. However, due to different detection principles, it is difficult to finely register the heterogeneous multisource data. In this section, we introduce sensor characteristics and system integration to analyze the sources of errors. Furthermore, the basis of the registration is analyzed to indicate the advantages of the proposed method.

A. Characteristics of Sensors

Hyperspectral scanners can simultaneously acquire images and continuous spectra of observed objects by capturing the electromagnetic solar irradiation reflected on the surface. Hyperspectral scanners have high spectral resolution acquisition capabilities, which capture an almost continuous spectrum in 400–2500 nm passively. In general, hyperspectral scanners have three scanning methods: whisk-broom, push-broom, and frame perspective. Among them, push-broom scanners are the preferred hyperspectral sensors for UAV platforms due to their high spatial resolution, small volume, and lightweight [36]. In this way, hyperspectral scanners acquire a 1-D array of across track per frame, and the spectral information is obtained through the spectroscopic system. The pixels along track are collected by the movement of the UAV platform to generate a complete hyperspectral image of one strip.

LiDAR is a piece of active remote sensing equipment that uses the laser as a light source and adopts photoelectric detection technology to obtain information. The LiDAR system includes a laser transmitter and a receiver. The ranging principle of LiDAR is to accurately measure the propagation time of a light pulse from emission to reflection back to the receiver. According to the known speed of light and the direction of laser emission, we can calculate the relative position of the scanning point to the LiDAR. Since the LiDAR is on a moving UAV platform, a POS is necessary to calculate the absolute position of the LiDAR data. POS consists of a global navigation satellite system (GNSS) and an inertial measurement unit (IMU), which can acquire the position and attitude information of the platform, respectively.

B. System Integration and Error Analysis

Considering the load of UAVs and the role of POS on hyperspectral scanners and LiDARs, the system adopts one POS to provide positioning information for multiple sensors in system integration. This integration method removes errors between different POSs and unifies the multisensor positioning reference. During the integration process, the hyperspectral scanner, LiDAR, and POS are fixed with rigid connections to reduce the instability of the integrated system. However, the position and orientation obtained by POS refer to the body frame, which is different from the hyperspectral scanner or LiDAR. Specifically, the spatial displacement between the POS body frame and the center of the sensor is defined as lever arm components, and boresight angles refer to the angles between the POS body frame and the scanner coordinate systems. It is necessary to calibrate the lever arm components

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and the boresight angles between sensors and POS, which determines the accuracy of the position and orientation of the sensors in the integrated system.

C. Basis of Registration

To align the heterogeneous data captured by the hyperspectral scanner and LiDAR, it is important to select the basis for registration. Intensity information and geometric information are usually considered for data registration.

In terms of intensity information, the hyperspectral scanner is designed to collect high-precision spectral intensity information of the observed object. Due to the influence of the atmospheric window, electromagnetic solar irradiation that contains a continuous spectrum is absorbed by atmospheric molecules, such as ozone, carbon dioxide, and water vapor. The signal-to-noise ratio (SNR) of the data acquired by the hyperspectral scanner is reduced in the atmospheric absorption bands [37]. As active detection equipment, LiDAR needs to consider more factors, such as laser technology, cost, and safety to human eyes. Furthermore, due to the limitations of the UAV platform, weight, volume, and detection distance of LiDAR also need to be considered. Currently, the commonly used LiDAR bands are mainly concentrated in the near infrared. LiDAR in these bands has mature technology, low cost, and small size, but the laser power is obviously limited due to the safety requirements of human eyes. The interference of solar electromagnetic radiation in this band will also affect the SNR of LiDAR intensity information [38]. The reliability of using the intensity information of hyperspectral and LiDAR for registration will be reduced under such circumstances.

For geometric information, the motion of the UAV platform and atmospheric disturbance in the stream layer lead to frequent changes in the attitude and position of the hyperspectral scanner, which causes obvious geometric distortion in the raw hyperspectral image. Compared with using GCPs to perform geometric correction on images acquired by push-broom hyperspectral scanners, direct geolocation geometric correction using POS demonstrates greater efficiency and accuracy [39]. The key aim of this method is to obtain the external parameters of the hyperspectral scanner and the coordinates and elevation information of ground points by aerial triangulation. However, different from frame sensors and three-line array sensors, single-line push-broom scanners cannot realize accurate mapping by aerial triangulation. The 3-D information detection capability of LiDAR provides complementary information for hyperspectral image geometric correction and the basis for stereo registration.

III. Method

The stereo registration method of hyperspectral images and LiDAR data proposed in this article can be divided into the following three parts:

1) multisource data preprocessing;
2) geometric correction of hyperspectral images and generation of true orthophoto images;
3) stereo registration of hyperspectral and LiDAR data.

A. Multisource Data Preprocessing

The original multisource data collected by the integrated system consist of three parts: hyperspectral raw data, LiDAR data, and attitude and position information. It is necessary to perform preprocessing steps on these original data before geometric registration to improve their data quality.

1) Raw Hyperspectral Data: The raw hyperspectral data are stored in the form of a data cube, and its three dimensions, respectively, represent the number of pixels, the number of frames, and the number of bands of push-broom imaging, without geographic coordinate information. Radiometric correction is the main step of raw hyperspectral data preprocessing, which includes radiometric calibration and atmospheric correction. Radiation calibration is to convert the DN value in the raw hyperspectral data into radiance by using the gain coefficient calibrated in the laboratory and the offset determined by the dark current collected after the flight experiment. Atmospheric correction is to convert the radiance into reflectivity through the standard reflector, and the purpose is to eliminate the errors caused by atmospheric scattering, atmospheric absorption, and atmospheric reflection.

2) POS: POS is composed of high-precision GNSS and IMU, which is one of the key parts of the integrated system for positioning. The measurement data of POS requires some necessary processing steps to improve the positioning accuracy of the GNSS and the attitude accuracy of the IMU. For the system error, we measure the level arm between the antenna and the IMU to correct the measured flight trajectory. Regarding the random error of the measurement, we take another GNSS base station on the ground as the reference station and use the carrier phase differencing technology to calculate with the GNSS on the UAV platform. Finally, this preprocessing step uses POSPAC based on the federated Kalman filter for GNSS and IMU to eliminate accumulated errors and improve accuracy.

3) LiDAR: The LiDAR data are the geometric reference for registration; some preprocessing steps are necessary to ensure the accuracy of the data. According to LiDAR data characteristics, the LiDAR system needs to be systematically calibrated using the LiDAR data collected from different airborne zones to eliminate the boresight angles between the LiDAR and IMU. Then, LiDAR data from different airborne zones are registered to reduce the spatial error of the LiDAR point cloud between flight strips. Denoising of the point cloud is also required to eliminate outliers caused by noise during LiDAR acquisition. Finally, considering the dispersion of point clouds, we use ENVI LiDAR to filter the point clouds to generate the DSMs, which are used to match the spatial features of hyperspectral images.

B. Geometric Correction of Hyperspectral Image and Generation of True Orthophoto Image

Considering the imaging mechanism of the push-broom hyperspectral scanner, the geometric correction of the hyperspectral image is realized by direct georeferencing based on the collinear equation, which describes the collinearity of the perspective center, the image point, and the corresponding

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object point of the scanner. According to the imaging principle and the collinear equation, we build a mathematical model as

$$\begin{bmatrix} X - X_S \\ Y - Y_S \\ Z - Z_S \end{bmatrix} = \lambda \mathbf{R}(\varphi, \omega, \kappa) \begin{bmatrix} x \\ y \\ -f \end{bmatrix} \tag{1}$$

where $P(X, Y, Z)$ represents the ground coordinates of the object point, (X_S, Y_S, Z_S) denotes the coordinates of the perspective center, λ is the scale factor, which depends on the focal length of the hyperspectral scanner and the distance between the object point and the perspective center, $p(x, y)$ represents the coordinates of the image point, f is the focal length of the hyperspectral scanner, and $\mathbf{R}(\varphi, \omega, \kappa)$ represents the rotation matrix determined by the exterior azimuth elements $(\varphi, \omega, \kappa)$, which denotes the roll, pitch, and yaw angles of the scanner, respectively. The specific expression is given as follows:

$$\mathbf{R}(\varphi, \omega, \kappa) = \mathbf{R}(\varphi) \cdot \mathbf{R}(\omega) \cdot \mathbf{R}(\kappa) \tag{2}$$

where

$$\mathbf{R}(\omega) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\omega & -\sin\omega \\ 0 & \sin\omega & \cos\omega \end{bmatrix} \tag{3}$$

$$\mathbf{R}(\varphi) = \begin{bmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{bmatrix} \tag{4}$$

$$\mathbf{R}(\kappa) = \begin{bmatrix} \cos\kappa & -\sin\kappa & 0 \\ \sin\kappa & \cos\kappa & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{5}$$

The external azimuth element angle of the push-broom hyperspectral imager can be obtained through the navigation angle provided by POS data. The conversion relationship between the rotation matrix of the navigation angle and external azimuth elements is given as follows:

$$\mathbf{R}(\varphi, \omega, \kappa) = \mathbf{C}_g^m \mathbf{C}_b^g(\Phi, \theta, \Psi) \mathbf{C}_c^b \mathbf{C}_i^c \tag{6}$$

where $\mathbf{C}_g^m$ represents the transformation matrix from the mapping coordinate system to the navigation coordinate system, $\mathbf{C}_b^g(\Phi, \theta, \Psi)$represents the transformation matrix from the navigation coordinate system to the IMU body frame coordinate system, (Φ, θ, Ψ) represents roll, pitch, and yaw obtained by POS, and $\mathbf{C}_c^b$ and $\mathbf{C}_i^c$ are the boresight rotation matrix and the transformation matrix of the sensor coordinate system to the image coordinate system.

Due to the lever arm between the POS and the hyperspectral sensor, the position measured by the POS is not the position of the perspective center. According to the measured value of the lever arm, the line elements of exterior orientation can be calculated by the following equation:

$$\begin{bmatrix} X_S \\ Y_S \\ Z_S \end{bmatrix} = \mathbf{C}_g^m \left(\begin{bmatrix} X_{\mathrm{IMU}} \\ Y_{\mathrm{IMU}} \\ Z_{\mathrm{IMU}} \end{bmatrix} + \mathbf{C}_b^g(\Phi, \theta, \Psi) \begin{bmatrix} X_l \\ Y_l \\ Z_l \end{bmatrix} \right) \tag{7}$$

where $[X_S, Y_S, Z_S]^{\mathrm{T}}$ represents the position of the sensor's perspective center, $[X_{\mathrm{IMU}}, Y_{\mathrm{IMU}}, Z_{\mathrm{IMU}}]^{\mathrm{T}}$ represents the position measured by POS, and $[X_l, Y_l, Z_l]^{\mathrm{T}}$ is the lever arm vector.

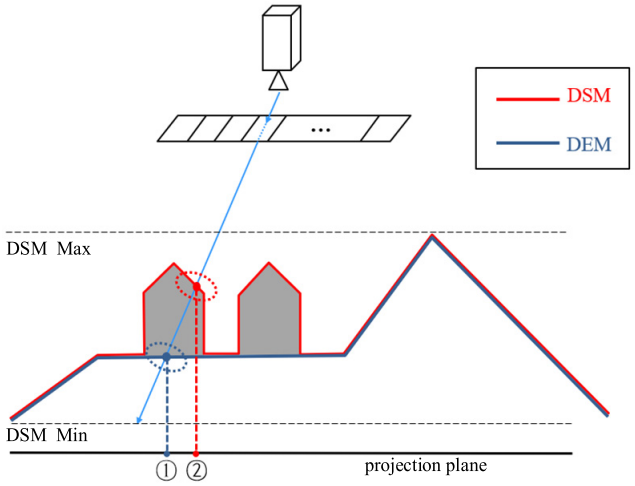


Fig. 2. Ray tracing module for orthorectification. The blue intersection 1 represents the projection position of the ray on the DSM, and the red intersection 2 represents the projection position of the ray on the DEM.

We use Z as a known quantity to solve for the scale factor and substitute it into (1)

$$\left.\begin{aligned} X &= X_S + (Z - Z_S)\frac{a_1 x + a_2 y - a_3 f}{c_1 x + c_2 y - c_3 f} \\ Y &= Y_S + (Z - Z_S)\frac{b_1 x + b_2 y - b_3 f}{c_1 x + c_2 y - c_3 f} \end{aligned}\right\} \tag{8}$$

where a_i, b_i, c_i represent the ith column elements of the first, second, and third rows of $\mathbf{R}(\varphi, \omega, \kappa)$, respectively.

The formula describes the position relationship between object points and image points. Since the push-broom hyperspectral scanner only obtains a 1-D array at a time, $x = 0$ in the heading direction. Y is determined by the pixel pitch. Normally, Z is provided by local average height or public DEM data for the geometric correction of hyperspectral images. However, for urban scenes and forestry scenes with complex height changes, high-precision DSM is required for orthorectification to correct projection errors caused by elevation changes.

As shown in Fig. 2, to eliminate the projection error and solve the double mapping problem, we use the ray tracing method based on the Z-buffer algorithm to orthorectify each pixel. This method is suitable for the integrated system of hyperspectral scanner and LiDAR, which determines the accurate position of the projection point of the ground object by establishing the virtual ray from the hyperspectral scanner to the surface of the object. With the help of the POS system, we can obtain the external orientation elements of each frame to determine the starting point and direction of the virtual ray. In order to improve the computational efficiency, we substitute the maximum and minimum height in DSM into the collinear equation, respectively, to reduce the calculation range of the intersection. The starting point and the endpoint of tracking can be specified as $[X_{\mathrm{ZMAX}}, Y_{\mathrm{ZMAX}}, Z_{\mathrm{ZMAX}}]$ and $[X_{\mathrm{ZMIN}}, Y_{\mathrm{ZMIN}}, Z_{\mathrm{ZMIN}}]$.

Since DSM is a rasterized processing result of the discrete LiDAR point cloud, we need to process the ray tracking

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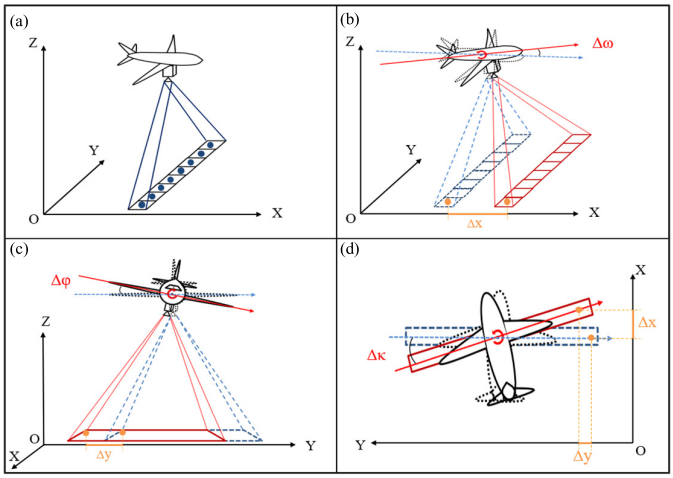


Fig. 3. Effects of boresight angles in different directions on the geographic location of hyperspectral images. It is assumed that the UAV flies along the X-axis. (a) No boresight angle bias. (b) Pitch boresight angle bias. (c) Roll boresight angle bias. (d) Yaw boresight angle bias.

direction when calculating the coordinates of the intersection point of the virtual ray and DSM. According to their plane coordinates in the projected coordinate system, we can calculate the slope and the step length of the ray in the X- and Y-directions per unit length. The tracking path is further obtained by the resolution of the DSM.

Based on the obtained path, we denote coordinates on the path as $[X_{\text{track}}, Y_{\text{track}}]$ in turn. Substitute $\text{DSM}(X_{\text{track}}, Y_{\text{track}})$ as Z into the collinear equation to calculate the assumed projection coordinates $[X_{\text{proj}}, Y_{\text{proj}}]$.

When the distance difference between the coordinates is smaller than the resolution of the DSM, the point is considered the intersection of the virtual ray and DSM. According to the principle of the Z-buffer algorithm, we compare the distances from all the intersection points of the virtual rays to the perspective center and define the coordinates of the intersection with the smallest distance as the orthorectified coordinates.

C. Stereo Registration of Hyperspectral and LiDAR Data

Since the calculation of the boresight angles is ignored in the orthorectification process in the previous part, there is still an obvious deviation between the orthorectified image and the LiDAR data. The unaligned DSM will cause geometric distortions in the orthorectification results of hyperspectral images. In this section, we analyze the image of the yaw angle to the registration error and utilize the stereoscopic information provided by LiDAR to establish a stereo registration method of hyperspectral image and LiDAR data, which is based on the virtual ray decomposition.

According to the LiDAR and the hyperspectral scanner use the same POS for positioning in the integrated system, the effect of random errors measured by the POS system can be ignored during the stereo registration process. System calibration of LiDAR has been implemented during the preprocessing process. Therefore, the registration error is mainly caused by the systematic error of the hyperspectral scanner. Since the internal parameters of the hyperspectral scanner have been strictly calibrated, and the level arm can be accurately measured, the boresight angles between the hyperspectral scanner and POS are the key issue in system calibration.

Based on the definition of the IMU coordinate axis in POS, boresight angles can be decomposed in three directions: roll, pitch, and yaw, which have different effects on the georeference of hyperspectral images. To briefly illustrate the effect of boresight angles on coordinate positioning, we assume that the attitude angles obtained by POS are 0. According to (1), the influence of boresight angles on georeferencing can be expressed as follows:

$$\begin{cases} \Delta X_i^{\text{pitch}} = 0 \\ \Delta Y_i^{\text{pitch}} = \lambda_i \sin(e_x) f \end{cases} \tag{9}$$

$$\begin{cases} \Delta X_i^{\text{roll}} = \lambda_i (\cos(e_y) x_i - \sin(e_y) f) \\ \Delta Y_i^{\text{roll}} = 0 \end{cases} \tag{10}$$

$$\begin{cases} \Delta X_i^{\text{yaw}} = \lambda_i \cos(e_z) x_i \\ \Delta Y_i^{\text{yaw}} = \lambda_i \sin(e_z) x_i \end{cases} \tag{11}$$

where e_x, e_y, e_z represent the boresight angles rotate on the X-, Y-, and X-axes, respectively, λ_i denotes the scale factor for the ith pixel, and x_i is the coordinates of the ith pixel in the sensor coordinate system.

The effect of viewing axis angles in different directions is shown in Fig. 3. According to the above analysis, we can summarize the following conclusions.

1) The bias of the boresight pitch angle affects the positioning of the pixels along the flight direction, which depends on the flight altitude, ground elevation, and angle size. Generally, the height of the flying platform is stable. When the terrain of the observation scene is flat, the coordinates of all pixels have the same change.

2) The bias of boresight roll angle affects the positioning of the pixels across the flight direction, which depends on the flight altitude, ground elevation, angle size, and pixel position. Affected by the central projection imaging method, the scale factor in the positioning equation of each pixel has different values. The change of each pixel coordinate is different when the bias of the boresight roll angle exists.

3) The bias of the boresight yaw angle affects the positioning of the pixels along the flight direction and across the flight direction at the same time. However, the coordinate change along the flight direction can be ignored compared with the vertical flight direction due to the small angle size. The impact on coordinates is related to flight altitude, ground elevation, and pixel location.

The boresight angles are often calibrated based on photogrammetric methods that require GCPs or many tie points between the hyperspectral images of adjacent flight strips to restore the elevation information in the image. These methods are more used for frame cameras because the imaging method of the push-broom of the hyperspectral scanner limits the accuracy of the elevation estimation. In order to avoid the influence of elevation error on boresight angle calibration, we propose a method for stereo registration of LiDAR and

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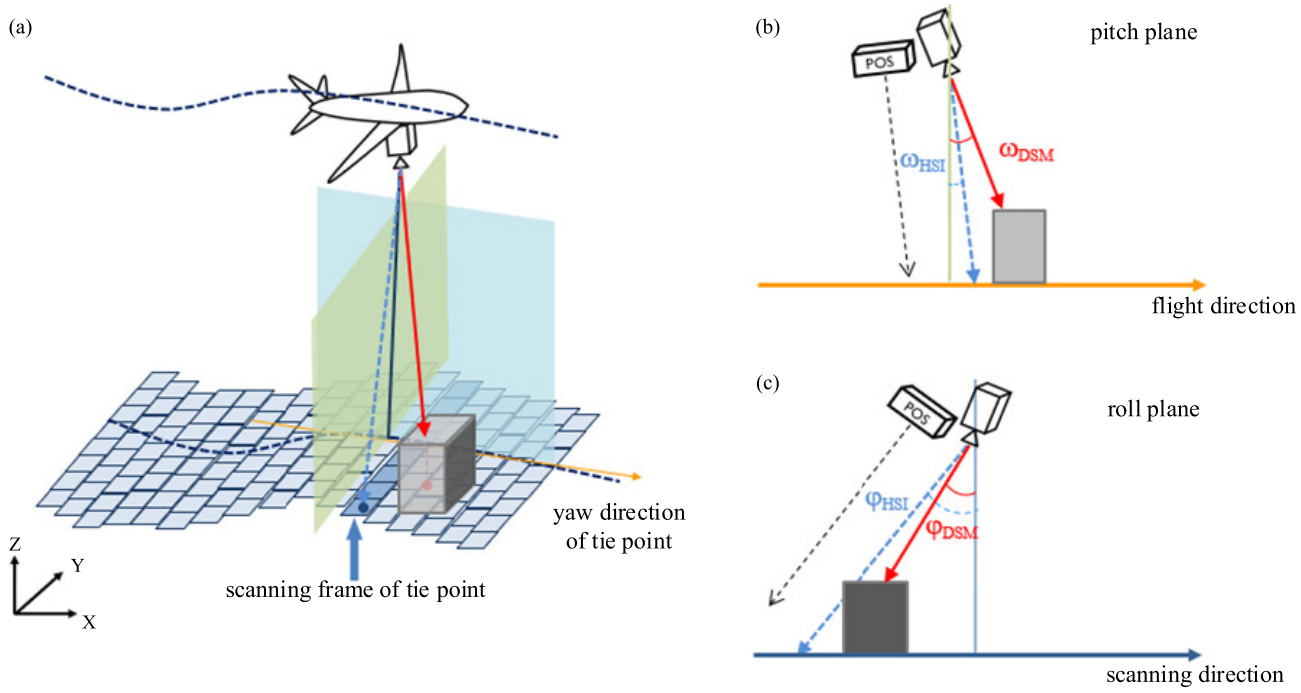


Fig. 4. Stereo registration based on virtual ray decomposition. The blue and red dots in (a) represent the conjugate feature points in hyperspectral and DSM, respectively, and the blue dotted line and red solid line are the constructed virtual rays. The green plane and the blue plane represent the roll and pitch planes decomposed according to the yaw direction, as shown in (b) and (c).

hyperspectral images based on elevation information in DSM and the virtual ray decomposition.

In this method, the pitch angle, the roll angle, and the yaw angle are calibrated separately. For the calibration of the yaw angle, we select the corresponding line features L_{dsm} in DSM and L_{hsi} in hyperspectral orthorectified images, which are perpendicular to the flight direction. Since the intensity value in DSM represents the altitude of the pixel location, the pixels in L_{dsm} have the same elevation. The pitch angle has the same effect on every pixel in L_{dsm}. According to formula (2), the roll angle affects each pixel differently, but the line feature perpendicular to the flight direction keeps its orientation unchanged from the roll angle. According to the definition of the yaw angle, the result of calibration can be expressed as follows:

$$\angle e_z = \arccos\left(\frac{\mathbf{l}_{\text{dsm}} \cdot \mathbf{l}_{\text{hsi}}}{|\mathbf{l}_{\text{dsm}}||\mathbf{l}_{\text{hsi}}|}\right) \tag{12}$$

where $\mathbf{l}_{\text{dsm}}$ and $\mathbf{l}_{\text{hsi}}$ represent the vectors from the start point to the end of the line feature in DSM and HSI, which can be calculated by the position of the line features.

According to the conclusions, the effects of roll and pitch angles on pixel coordinates are orthogonal to each other. After the yaw angle is estimated, we bring the boresight angle matrix of the yaw angle into the formula to reposition the hyperspectral image. For the calibration of row and pitch angles, we established virtual rays of feature points and decomposed them according to the imaging model during the flight of the UAV, as shown in Fig. 4. The conjugate feature points p_{dsm} in the DSM image and p_{hsi} in the hyperspectral image are selected as the endpoints of the virtual rays. The starting point of the virtual ray is the perspective center position of the frame containing p_{hsi}, which can be obtained by calculating the external orientation elements. Since the flight direction has been calibrated, the virtual rays can be decomposed along and across the flight direction. Based on the virtual ray of the DSM feature point, the bias of the boresight pitch angle and the bias of boresight roll angle can be calculated

$$\angle e_x = \arccos\left(\frac{\mathbf{r}_{\text{dsm}}^{\text{roll}} \cdot \mathbf{n}}{|\mathbf{r}_{\text{dsm}}^{\text{roll}}|}\right) - \arccos\left(\frac{\mathbf{r}_{\text{hsi}}^{\text{roll}} \cdot \mathbf{n}}{|\mathbf{r}_{\text{hsi}}^{\text{roll}}|}\right) \tag{13}$$

$$\angle e_y = \arccos\left(\frac{\mathbf{r}_{\text{dsm}}^{\text{pitch}} \cdot \mathbf{n}}{|\mathbf{r}_{\text{dsm}}^{\text{pitch}}|}\right) - \arccos\left(\frac{\mathbf{r}_{\text{hsi}}^{\text{pitch}} \cdot \mathbf{n}}{|\mathbf{r}_{\text{hsi}}^{\text{pitch}}|}\right) \tag{14}$$

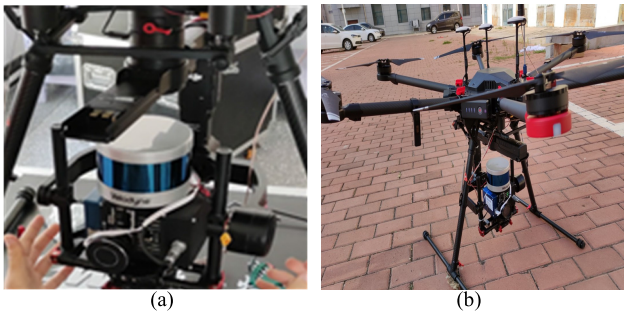
where $\mathbf{r}_{\text{dsm}}^{\text{pitch}}$ and $\mathbf{r}_{\text{dsm}}^{\text{roll}}$ represent vectors of the virtual ray of DSM projections on pitch plane and roll plane, $\mathbf{r}_{\text{hsi}}^{\text{pitch}}$ and $\mathbf{r}_{\text{hsi}}^{\text{roll}}$ denote vectors of the virtual ray of HSI projection results in the same way, and $\mathbf{n}$ represents the unit normal vector vertically downward.

IV. Experimental Results and Discussion

A. Data Acquisition System

The data acquisition system is mainly composed of a hyperspectral scanner, LiDAR, POS, and UAV platform, as shown in Fig. 5. The integrated system uses the Headwall Nano-Hyperspec [40] push-broom scanner as a hyperspectral sensor, which is specially designed for UAV platforms with ideal size, weight, and power consumption. It contains aberration-corrected optics, an internal solid-state drive (SSD), and lenses. LiDAR data are acquired by Velodyne Puck LITE [41], which is improved from VLP-16. It has the

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(a) (b)

Fig. 5. Data Acquisition System. (a) Multisensor (Velodyne Puck LITE, Headwall Nano-Hyperspec, and APX-15) integrated system. (b) DJI Matrice 600 Pro UAV platform.

TABLE I
MAIN PARAMETERS OF THE SENSORS

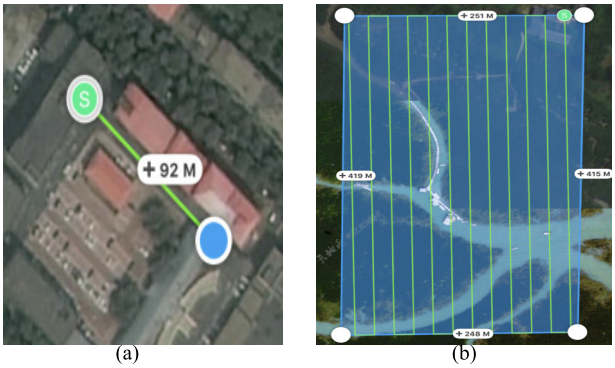
Sensors	LiDAR	Hyperspectral scanner
Type	Velodyne Puck LITE	Headwall Nano-Hyperspec
Principle	active	passive
Detection mode	Whisk-broom	Push-broom
Weight	590g	783g
Wavelength range	903nm	400-1000nm
Bit depth	8bits	12bits
Detector pixel pitch	——	7.4μm
Measurement Range	100m	——
Field of View (°)	across track: 360° along track: +15° to -15° (30°)	across track: 0.053° along track: 0.053°

same performance and lighter weight as VLP-16. The specific parameters of the two sensors are shown in Table I. The system integrates Trimble APX-15 [42] to provide attitude and position information for sensors and uses POSPAC for postprocessing of the raw data. For the UAV platform, DJI Matrice 600 Pro is chosen to integrate the hyperspectral scanner and LiDAR. Moreover, a DJI Ronin MX gimbal stabilizer was used to stabilize the attitude of the sensors.

B. Data Description

In this study, datasets of two typical scenarios were acquired for experiments, which include an urban scene of the Harbin Institute of Technology and a forestry scene of Shankou Town. The datasets contain raw hyperspectral data, LiDAR data, POS data, and GNSS base station data.

1) Harbin Institute of Technology: This dataset was captured at the Harbin Institute of Technology, Harbin, Heilongjiang, China, in 2021. The flying height is 80 m, and



(a) (b)

Fig. 6. Flight route planning and satellite images of experimental scenarios. (a) Harbin Institute of Technology. (b) Shankou Town.

the ground resolution of the hyperspectral image is about 0.1 m. The LiDAR point cloud density is 100 pts/m^2 on average. The observation scene is an urban scene, which mainly includes features such as parking lots, vehicles, buildings, and trees. Affected by artificial buildings and vehicles, the scene elevation changes are complex, and the range of changes exceeds 30 m. Fig. 6(a) shows flight route planning and satellite images in this dataset.

2) Shankou Town: This dataset was collected in Shankou, Beihai, Guangxi, China, in 2021. The flying height is 80 m, and the flying speed is 7 m/s. The dataset consists of 12 flight strips. The main features of the observation scene include mangroves, trestles, and tidal ditches. Mangroves are densely distributed, and the observation scene is uneven. Fig. 6(b) shows flight route planning and satellite images of this dataset.

C. Results and Analysis

To verify the effectiveness of the proposed method, we compare our method with the frame camera-improved method [43] and dedicated software SpectralView [44] on the two datasets. RGB frame images are commonly used to improve the quality of hyperspectral orthorectification due to the lower acquisition cost and 3-D reconstruction capability of frame cameras. SpectralView is the dedicated software for headwall hyperspectral scanners, which is used by several studies to generate or display results due to the ability of hyperspectral image orthorectification and fusion with LiDAR data [26], [43]. Its mechanism is to adjust each frame to a specific known GPS position and UAV orientation.

Noted that we experimentally found that intensity- or feature-based methods are not suitable for comparison due to the low SNR of the intensity information of UAV LiDAR data.

We did not report the experimental results of intensity- or feature-based methods due to their poor performance. This attributes to the fact that the intensity information for UAV LiDAR data exhibits very low SNR [38].

The visualization of dataset one is shown in Fig. 7. The effect of each step in our method is demonstrated by the results of the single strip in the first dataset, as shown in Fig. 8. In addition, experiments on the second dataset containing multiple strips are carried out to show the applicability of the proposed method to different scenarios and the registration

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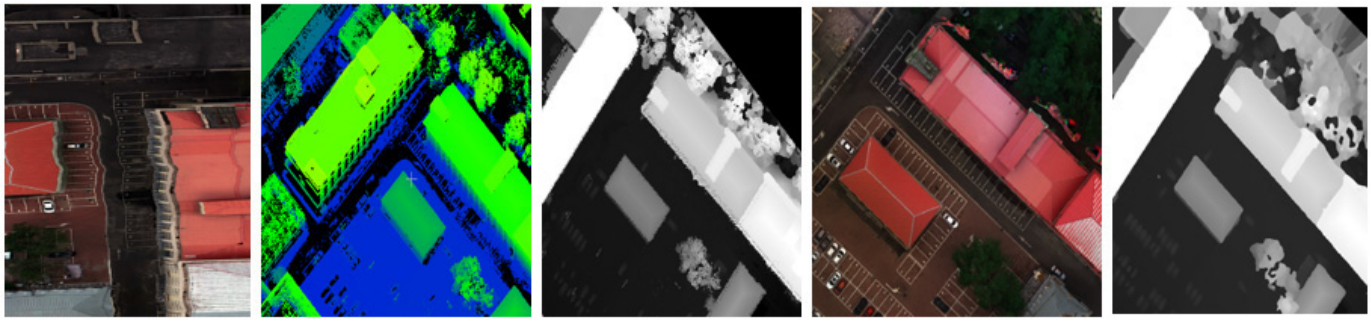


Fig. 7. Dataset of the Harbin Institute of Technology. (a) HSI. (b) LiDAR data. (c) DSM by LiDAR. (d) Mosaic of the frame camera. (e) DSM by the frame camera.

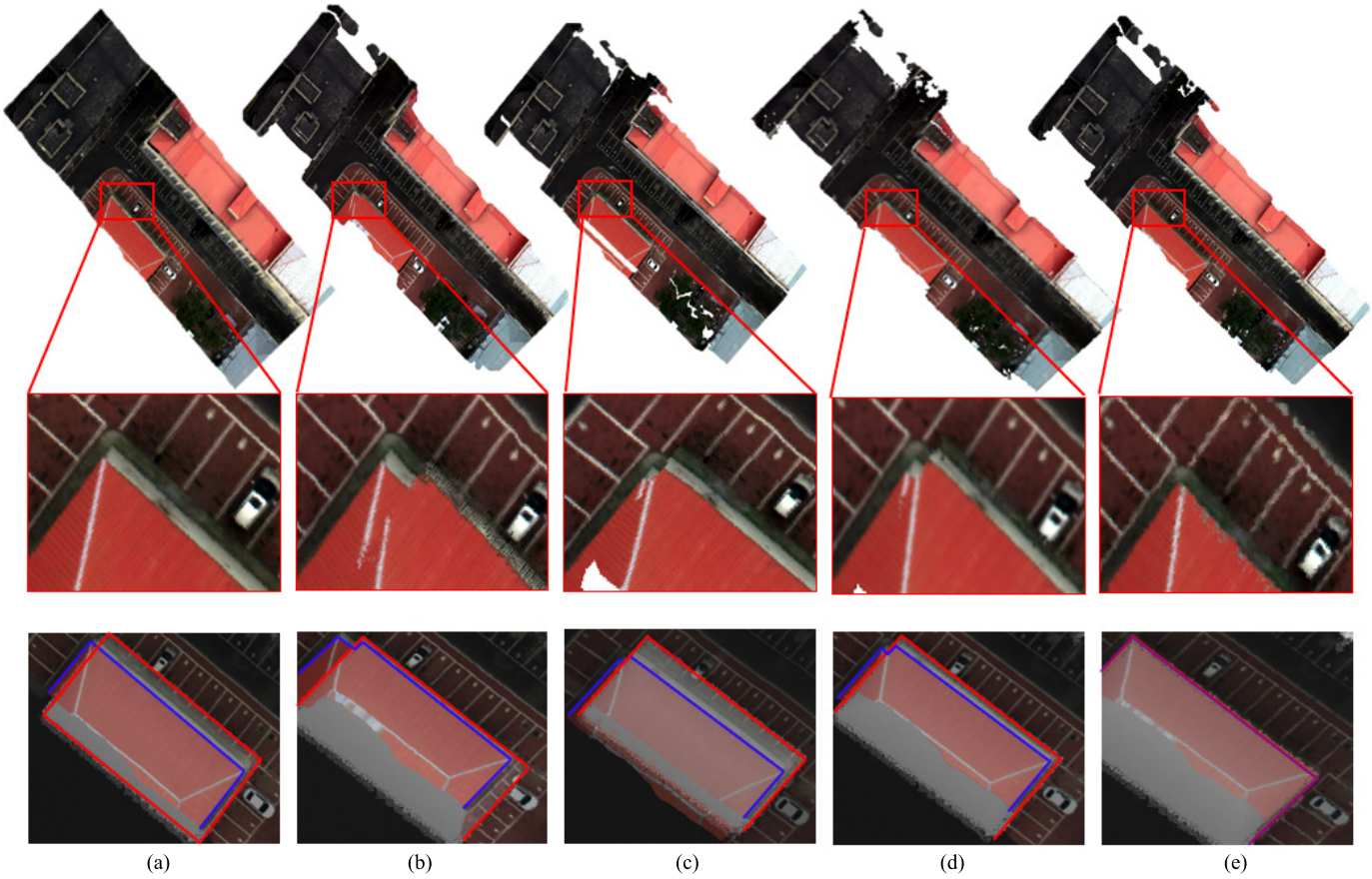


Fig. 8. Registration results of the Harbin Institute of Technology dataset. From (Top) to (Bottom) are the hyperspectral image, the partially zoomed-in image, the hyperspectral image, and the LiDAR DSM overlay display. To clearly show the registration effect, we use blue lines to represent the edges of buildings in the hyperspectral image and red lines to represent the edges of buildings in DSM. (a) DEM-based. (b) DSM-based. (c) camera-improved. (d) SpectralView. (e) Stereo Registration.

effect for adjacent flight strips. Due to the large number of flight strips and the similar visual effects of the intermediate results, Fig. 9 merely shows the mosaic after registration. We evaluate the experimental results through visual effects and quantitative evaluation to demonstrate the effect of the proposed method.

1) Qualitative Analysis: It can be seen from the registration results in Fig. 8 that our method has better visual effects than the comparison methods. We use blue lines to represent the edges of buildings in the hyperspectral image and red lines to represent the edges of buildings in DSM to better show the registration results of the edges. Compared with the frame camera-improved method, our method's advantages are mainly reflected in two aspects. First, the results of hyperspectral geometric correction are affected by the accuracy of DSM. Although the 3-D reconstruction of the frame image can obtain the relative height information in the image according to the photogrammetry, the accuracy of the height is lower

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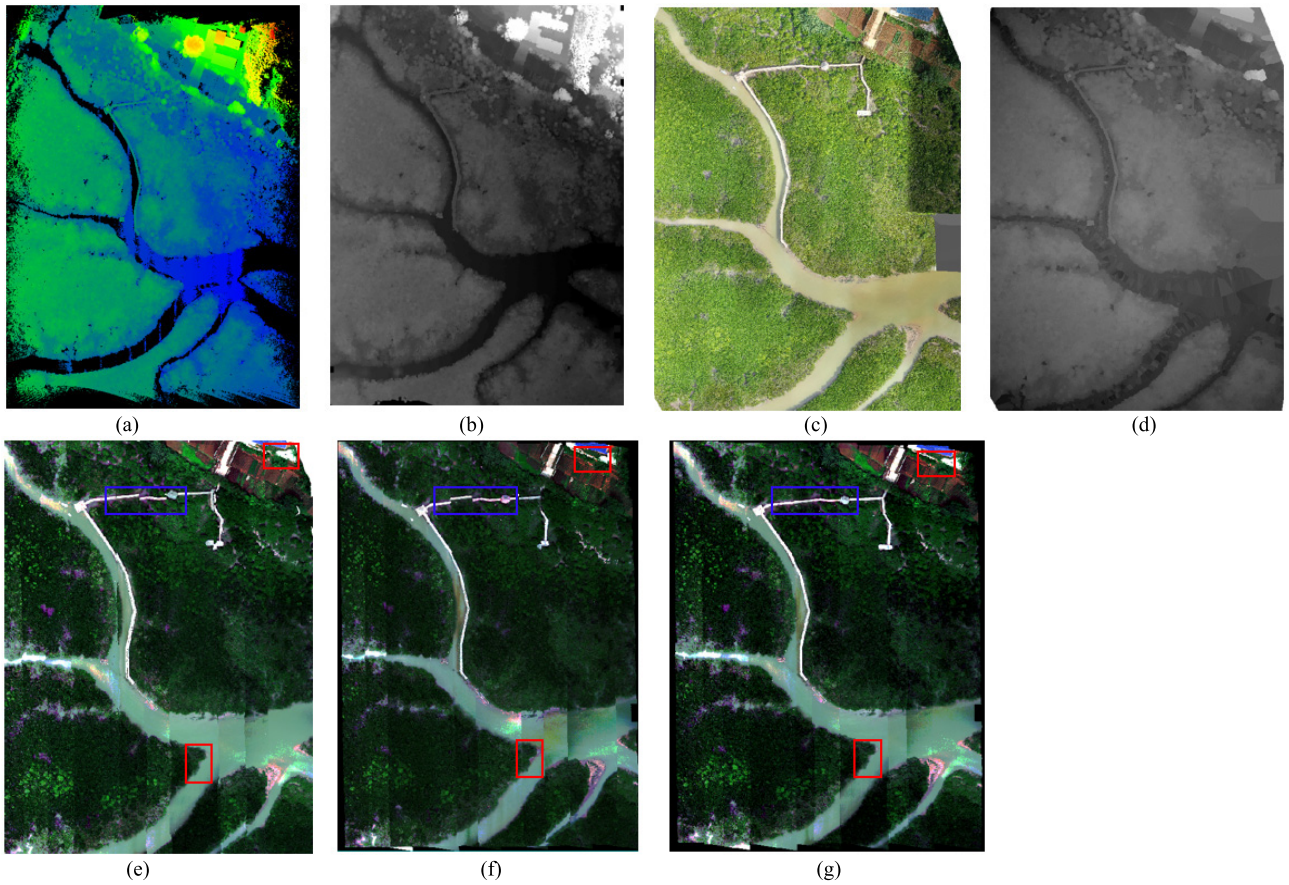


Fig. 9. Shankou Town dataset and experimental registration results. (a) LiDAR point cloud. (b) DSM by LDAR. (c) Mosaic of frame camera. (d) DSM by frame camera. (e) Frame camera improved. (f) SpectralView result. (g) Our method.

than that of the LiDAR, in the case of no GCPs. Second, methods of using frame images to improve the orthophoto correction quality of hyperspectral images require relatively flat terrain [43]. It is difficult to achieve ideal results in scenes with obvious elevation changes. The dedicated software SpectralView has achieved relatively better results, but its process involves manual iterative adjustment and requires analysis of the results compared to a Google Earth or Google Maps view to obtain the calibration parameters. The image processing results corresponding to the calibration parameters are not completely aligned with the LiDAR data. Our method achieves the best alignment at the edges of the elevation changes without using control point and intensity information.

The effectiveness of each part of the proposed method is also shown in Fig. 8. We can obtain the geometric correction result through the position and posture information provided by POS, as shown in Fig. 8(a). The correction process eliminates the obvious geometric distortion in the hyperspectral image and gives the image geographic location information. It can be seen from the overlay display with DSM that there are obvious positioning errors, such as buildings and vehicles in the image affected by the projection mode. Fig. 8(b) shows hyperspectral orthophoto based on DSM. The projection error caused by elevation is significantly reduced. The aspect ratio of buildings in hyperspectral images is close to that in real scenes. DSM contains the elevation information of ground objects, which is important for accurate orthorectification of hyperspectral images. However, the hyperspectral image is still not registered with the DSM due to the boresight angles, which leads to incorrect DSM-based hyperspectral orthorectification, such as the misalignment of the building edge. The results in Fig. 8(e) show that the proposed stereo registration method aligns the heterogeneous data effectively. Partially zoomed-in images show detail on the edge of the house. The overlay display of the image and the DSM demonstrates the registration performance of the proposed method.

The experimental results in Fig. 9 show that our method still achieves the best visual results on the second dataset, demonstrating its applicability to different scenarios. Fig. 9(e)–(g) shows the mosaic results of contrast methods and our method, respectively. It is clearly seen from the edge areas of the trestle bridge (within the blue box) and the edge of the mangrove that our registration result is less misaligned at the junction of the flight strips. In Fig. 10, the southern and northern parts of the hyperspectral image (within the red boxes) are selected to be enlarged to illustrate the impact of the global registration effect. In addition, from the results shown in the overlay, the registration performance with DSM is confirmed. The edge of the hyperspectral image and the edge of the LiDAR DSM are represented by different colored lines. In our results, the hyperspectral image and DSM are more closely aligned at the edge of the mangrove.

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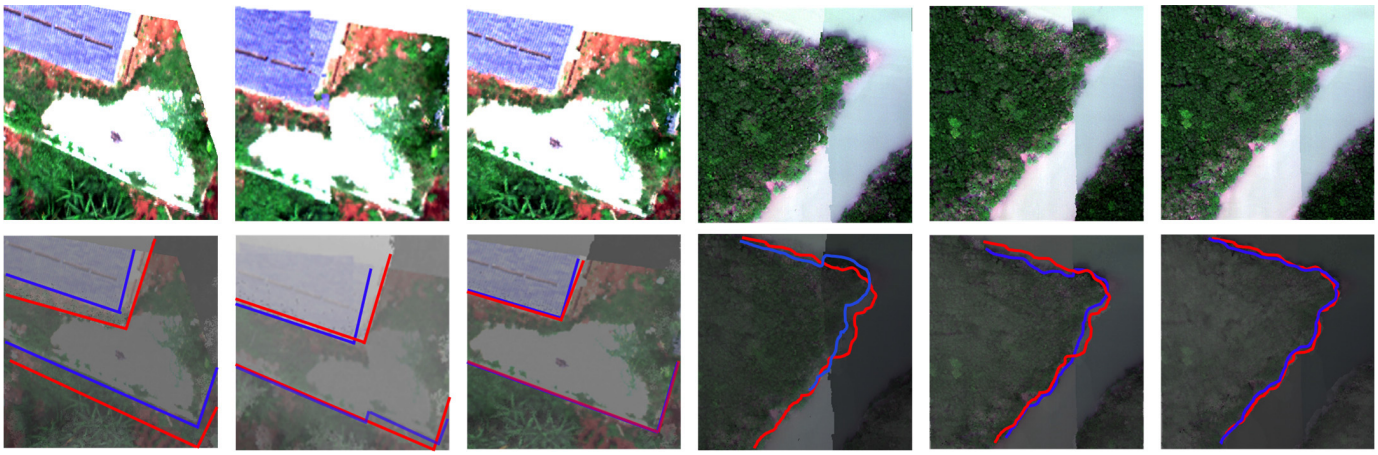


Fig. 10. Zoomed-in areas of Shankou Town experimental registration results. (a)–(c) Results of frame camera improved method, SpectralView, and our method in Fig. 9 (top red box), respectively. (d)–(f) Results in Fig. 9 (bottom red box). The results in the first row are the hyperspectral images after registration, and the results in the second row are the overlay display images. Blue lines represent partial edges in hyperspectral images, and red lines represent edges in DSM.

TABLE II
RMSE OF THE DIFFERENCE BETWEEN THE TIE POINTS' GROUND COORDINATES AFTER DIFFERENT REGISTRATION STEPS

Methods	RMSE of X coordinate residuals (m)	RMSE of Y coordinate residuals (m)	RMSE of tie points plan distance residuals (m)
DEM-based	2.7478	3.0985	4.1414
DSM-based	1.6081	1.1532	1.9788
Camera-improved	2.0673	1.9918	2.8707
SpectralView of Headwall	1.0423	1.2481	1.6261
Stereo registration of our method	**0.4440**	**0.2844**	**0.5273**

It is worth emphasizing that no registration process is performed on hyperspectral images between different flight bands in this work. Considering the significant influence of surface fluctuation on hyperspectral image correction, our method achieves good registration between strips by registering the hyperspectral image of each flight strip and DSM.

2) Quantitative Analysis: The RMSE between manually selected tie points is used to evaluate the registration accuracy of hyperspectral images and LiDAR data. In order to obtain accurate evaluation results, the tie points are selected manually and evenly distributed in the whole image and laid image control point markers for auxiliary selection. Since the registration process is more concerned with the relative position relationship rather than the absolute positioning accuracy, commonly used quantitative evaluation metrics include similarity measures of image intensity information, error calculations using all or uniformly sampled 3-D point clouds, and the root mean square error (RMSE) obtained by tie points [11], [45], [46]. Due to the SNR of LiDAR data and the dimensional difference of heterogeneous data, the first two evaluation methods are not applicable in this study.

Table II lists the RMSE of the differences between tie points in the first dataset. In our method, the ray tracing model is used to generate the orthophoto of the hyperspectral image. The elevation information in the hyperspectral image is completely derived from LiDAR, so the deviation in the Z-direction has no reference significance. It can be seen that the RMSE of the differences between tie points in the XY-direction gradually decreases through a series of processing steps. Affected by the change of elevation, the orthorectification improved by the frame camera has a larger error than the methods based on LiDAR. Compared with the geometrically corrected hyperspectral image generated based on DEM, the true orthophoto image generated using LiDAR data is closer to LiDAR data. However, this registration error is still at the meter level, which is mainly caused by the boresight angles between the hyperspectral scanner and POS. After stereo registration, the RMSE of the differences in the XY-direction (m) is reduced to 0.5273 m.

According to the accuracy of the sensors, the localization performance of the POS, and the experimental conditions, this registration error in the urban scene is acceptable. In addition, the main factor leading to the error is the smoothness of the DSM edge, which is caused by the filtering of LiDAR in the generation process. DSM exists some errors in the position of elevation change. Since the corresponding feature point and

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TABLE III
RMSE OF THE DIFFERENCE BETWEEN THE TIE POINTS' GROUND COORDINATES FOR SHANKOU TOWN DATASET

Sortie	flight strip	RMSE of X coordinate residuals (m)			RMSE of Y coordinate residuals (m)			RMSE of tie points plan distance residuals (m)		
		Camera-improved	SpectralView	Ours	Camera-improved	SpectralView	Ours	Camera-improved	SpectralView	Ours
1	1	1.8371	2.0870	**0.7208**	2.0891	1.8542	**0.1631**	2.7459	2.7449	**0.5839**
	2	2.4008	1.8414	**0.4218**	1.9672	1.5179	**0.1419**	3.1039	2.3916	**0.4450**
	3	1.9664	2.2716	**0.6557**	1.7391	1.7558	**0.1712**	2.6251	2.8711	**0.6777**
	4	1.4951	1.7035	**0.3171**	1.7558	1.5924	**0.2760**	2.3061	2.3340	**0.4204**
2	1	2.0972	2.3026	**0.7856**	1.9179	2.4713	**0.6793**	2.8419	3.3778	**1.0386**
	2	1.6727	1.1913	**0.3371**	1.8779	1.2942	**0.3112**	2.5148	1.7645	**0.4588**
	3	1.4199	1.7391	**0.5285**	1.0550	1.5497	**0.2630**	1.7689	2.3228	**0.5903**
	4	1.4925	1.2054	**0.2290**	1.1558	1.2518	**0.5383**	1.8877	1.7378	**0.5850**
3	1	0.8575	0.5176	**0.3378**	2.2716	2.8383	**0.7394**	2.8711	2.8851	**0.8129**
	2	1.1413	1.2262	**0.4909**	1.7035	1.7847	**0.3377**	2.0505	2.1653	**0.5958**
	3	2.3375	1.8085	**0.3532**	1.5924	1.2725	**0.3685**	2.8284	2.2113	**0.5104**
	4	1.4971	1.3765	**0.3063**	0.8926	0.7119	**0.2077**	1.7430	1.5497	**0.3701**

TABLE IV
RMSE OF THE DIFFERENCE BETWEEN THE TIE POINTS' GROUND COORDINATES IN ADJACENT HYPERSPECTRAL BANDS AFTER REGISTRATION

Sortie	RMSE of X coordinate residuals (m)			RMSE of Y coordinate residuals (m)			RMSE of tie points plan distance residuals (m)		
	Camera-improved	SpectralView	Ours	Camera-improved	SpectralView	Ours	Camera-improved	SpectralView	Ours
1	2.8725	2.9672	**0.4463**	2.8947	2.9681	**0.1640**	4.0780	4.1969	**0.4755**
2	2.8715	2.0972	**0.3922**	1.4561	2.5222	**0.1712**	3.2196	3.2802	**0.4280**
3	2.8785	2.7878	**0.4387**	2.3766	2.7516	**0.2238**	3.7328	3.9170	**0.4925**

most of the tie points are the corners in the DSM, the selection of matching points in the DSM leads to some registration errors and evaluation errors.

Table III shows the experimental evaluation results in the second dataset. Multiple sets of experiments in mangrove scenes prove that the proposed method is effective for nonurban scenes, and the registration performance is not limited by sorties and flight strips. The error in the results of the frame-camera improved method is reduced on this dataset, which is close to the SpectralView results. This is due to the reduced elevation change in the forest relative to the urban scene. Compared with SpectralView, the RMSE differences between tie points in the *XY*-direction are significantly reduced. Its mean value decreased to 0.5907 m. The standard deviation of the registration error between the flight strips is 0.1860 m, which can indicate the stability of the registration method. As can be seen from Table IV, after the stereo registration of each hyperspectral image with LiDAR data, the RMSE of the differences between tie points in adjacent flight strips is reduced to 0.4661 m. It is shown that the proposed method can realize the registration of hyperspectral images obtained by different flight strips simultaneously.

TABLE V
COMPARISON OF EXECUTION TIME AND MEMORY USAGE

Area	Methods	Execution time	Memory usage
HIT	Camera-improved	16.64s	136MB
	SpectralView	9.10s	98MB
	Ours	**18.24s**	**129MB**
Shankou	Camera-improved	34.82s	287MB
	SpectralView	24.68s	226MB
	Ours	**46.24s**	**251MB**

The results in Table V provide a comparative analysis of the computational cost of the proposed method and the comparison methods. The execution time and memory usage are reported for the two datasets. All experiments were run on the same personal computer with CPU: Intel i7 9800X, GPU:

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Nvidia GeForce GTX 1660, and 128 GB RAM. The results in Table V show that our method yields longer execution time and requires more memory compared to the reference methods, but it is in an acceptable range and obtains higher accuracy results.

V. Conclusion

In this article, a new stereo registration method of push-broom hyperspectral scanner and LiDAR on UAV platforms is proposed, which aligns hyperspectral images and LiDAR data without intensity information. The proposed method is based on the physical model of the integrated system consisting of a push-broom hyperspectral scanner and LiDAR. Experiments on two datasets from different scenarios demonstrate the effectiveness and robustness of this method in visual and quantitative evaluations.

According to the analysis of the source of the registration error, the registration process first preprocesses the data to ensure the geographic accuracy of the LiDAR data and takes the DSM generated by LiDAR data as the reference of registration. Then, with the assistance of DSM, the collinear equation is established according to the push-broom imaging model, and the ray tracing method is used to orthorectify the hyperspectral images, eliminating projection errors caused by elevation. Finally, the boresight error that causes the misalignment of hyperspectral images and LiDAR data is corrected by the virtual ray decomposition model. Different from the existing registration methods, the registration process does not rely on the intensity information of the UAV airborne LiDAR and GCPs, which makes our method more reliable and wider applicability.

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Yanfeng Gu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined the School of Electronics and Information Engineering, HIT, as a Lecturer. He was appointed as an Associate Professor at HIT in 2006; meanwhile, he was enrolled in the first Outstanding Young Teacher Training Program of HIT. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California at Berkeley, Berkeley, CA, USA. He is currently a Professor with the Department of Information Engineering, HIT. He has published more than 100 peer-reviewed papers and four book chapters. He is the inventor or a co-inventor of 20 patents. His research interests include space-intelligent remote sensing and information processing, multimodal hyperspectral remote sensing, and spaceborne time-series image processing.



Chen Wang (Graduate Student Member, IEEE) received the B.E. degree in electronics and information engineering from the Harbin Institute of Technology, Harbin, China, in 2019, where he is currently pursuing the Ph.D. degree in information and communication engineering.

His research interests include hyperspectral images and light detection and ranging (LiDAR) data processing and its application in remote sensing.



Xian Li (Member, IEEE) received the Ph.D. degree in instrument science and technology from the Harbin Institute of Technology (HIT), Harbin, China, in 2021.

From 2018 to 2020, he was a Doctoral Researcher with the Department of Telecommunications and Information Processing, Ghent University, Ghent, Belgium, supported by the China Scholarship Council. He is currently an Assistant Professor with the School of Electronics and Information Engineering, HIT. His research interests include deep learning, hyperspectral remote sensing, and image analysis.

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